Differences In Dynamic Reconfiguration Of Whole-Brain Connectivity Are Related To Individual Differences In Working Memory Task Performance

Maren Wehrheim[1,2], Christian J. Fiebach[1,3]

[1] Goethe University, [2] Frankfurt Institute for Advanced Studies, [3] Brain Imaging Center, Goethe University

Introduction

What are the neural underpinnings of working memory?

- Working memory performance underlies communication between large scale functional networks [1,2]
- Evidence for increased as well as decreased connectivity strength due to working memory load [3,4]

Dynamic connectivity

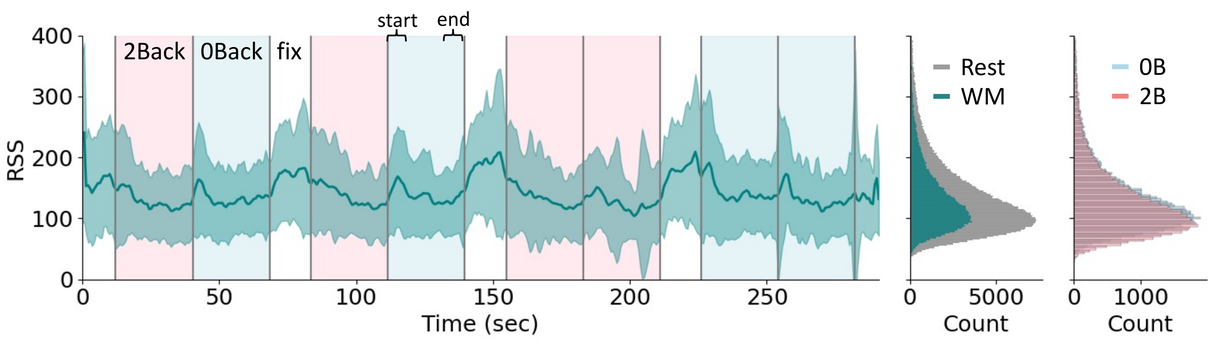
- Changes in dynamic connectivity have been observed within different task domains [5,6] and neurodevelopmental and psychotic disorders [7-9]
- Higher network flexibility was associated with higher working memory performance [10]

Research questions

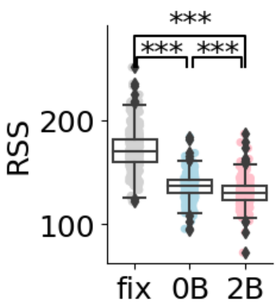
I. Do tasks with higher cognitive load require lower or higher connectivity?
II. Are dynamic changes in co-activation during tasks related to individual differences in working memory performance?

Results I: dynamic changes in task-related co-activation

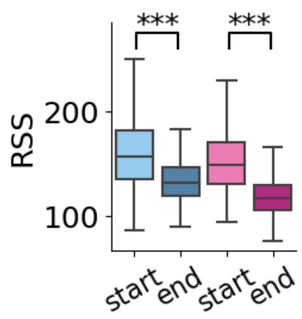
Lower connectivity during high than low load tasks



- 0B & 2B temporal connectivity appear similar
- Similar distribution of co-activation during task and rest

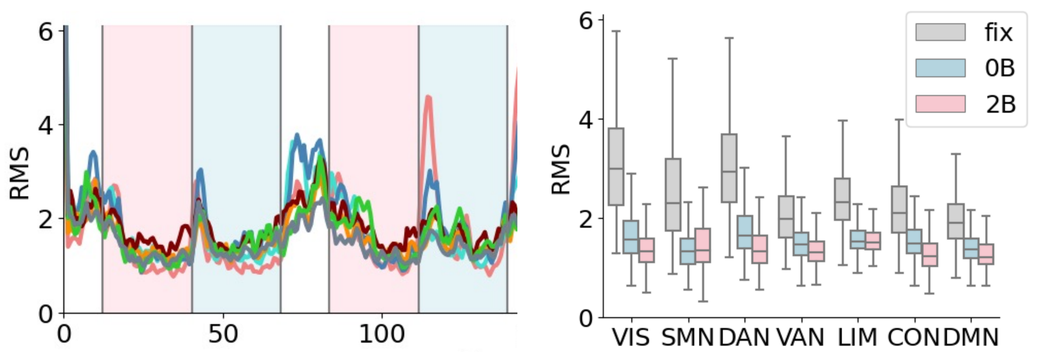


Lowest co-activation during 2B



Decreasing co-activation during task blocks

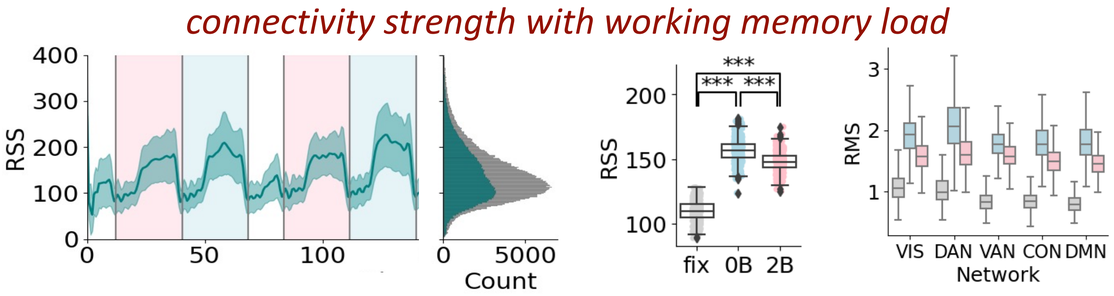
No differences between functional networks



Visual (VIS), Somatomoter (SMN), Dorsal attention (DAN), Ventral attention (VAN), Limbic (LIM), Control (CON), Default mode (DMN)

Controlling for task-activation effects

Change in pattern of temporal dynamics but same reduction in connectivity strength with working memory load

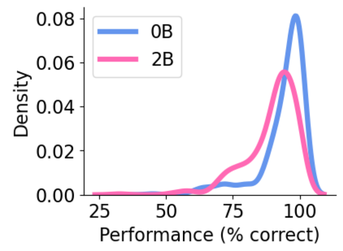


Methods

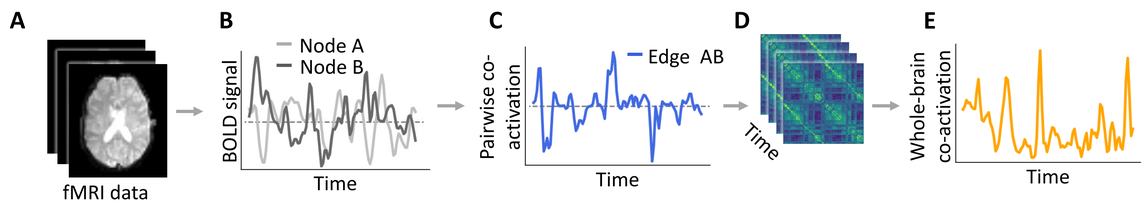
Data

- Human Connectome Project (HCP [11]; N=331 unrelated subjects)
- 0-back and 2-back block task
- 4 blocks (25sec) per load condition: faces, body parts, places, tools

Behavioural measure: Task performance over all blocks



Cofluctuation time series

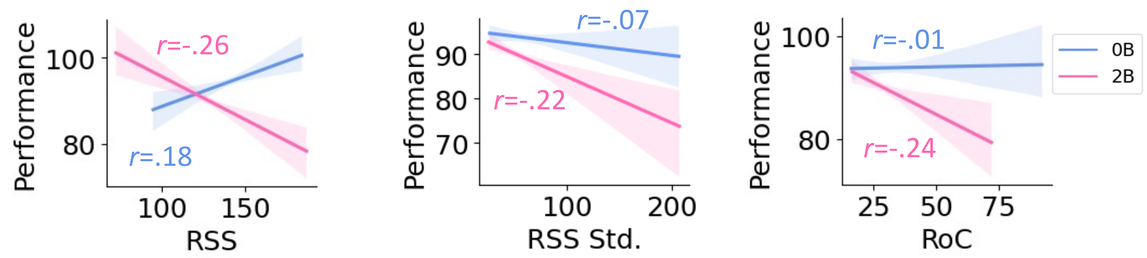


- Parcellate fMRI data (**A**) into 200 Schaefer nodes [12] (**B**)
- Compute instantaneous co-activation (**C**) between all brain nodes (**D**)
- Compute whole-brain co-activation as root-sum square between all edges (**E**)

Dynamic measures

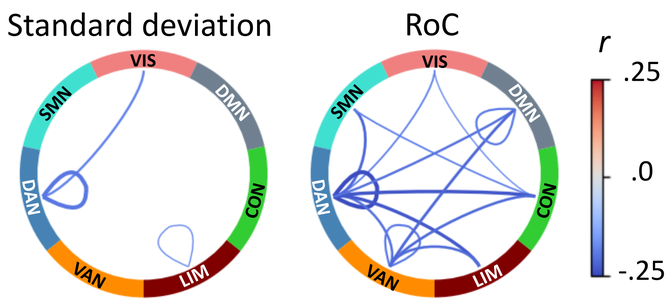
- Variability within complete task block measured as *standard deviation (Std.)*
- Moment to moment variability measured as *rate of change (RoC):* $\sum_{t}^{T-1} x_{t+1} - x_t$, where x_t is functional activity at time t

Results II – dynamic co-activation & individual differences in working memory



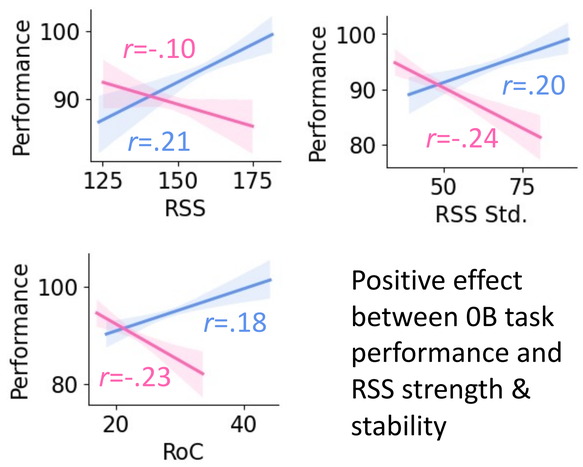
- Lower RSS and variability → higher 2B task performance
- Higher RSS → higher 0B task performance

Attention and control networks relevant for 2B task performance



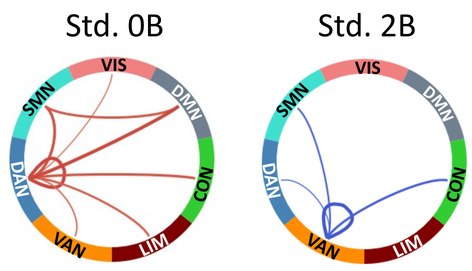
Rate of change in attention and control network co-activations best explain individual differences in 2B task performance

Controlling for task-activation effects



Positive effect between 0B task performance and RSS strength & stability

Higher (lower) standard deviation in dorsal (ventral) attention network correlates with 0B (2B) task performance



Conclusion

- Tasks with higher working memory load require less whole-brain connectivity than lower load tasks
- Temporal stability of task-related functional connectivity within attention networks is related to individual differences in working memory performance

Resources

[1] Cohen & D'Esposito (2016), *Journal of Neuroscience*
[2] Avery et al. (2020), *Journal of cognitive neuroscience*
[3] Ginset et al. (2011), *NeuroImage*
[4] Liang et al. (2013), *PNAS*
[5] Liu et al. (2018), *Human brain mapping*
[6] Bassett et al. (2011), *PNAS*
[7] Cui et al. (2021), *Cerebral Cortex*
[8] He et al. (2021), *Frontiers in Psychiatry*
[9] Xie et al. (2021), *bioRxiv*
[10] Braun et al. (2015), *PNAS*
[11] Van Essen et al. (2013), *NeuroImage*
[12] Schaefer et al. (2018), *Cerebral Cortex*